

A Conditional Substrate Localisation of the Hilbert–Pólya Problem

A target specification on an infinite-dimensional lift of the τ -fixed block of V_{600} : if a generator with the stated substrate constraints exists, its positive eigenvalues in multiplicity-respecting bijection with ζ 's upper-half-plane non-trivial zeros, then RH follows; the converse is the named open problem

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Abstract

The Riemann Hypothesis (RH) is the conjecture that every non-trivial zero of $\zeta(s)$ has $\text{Re}(s) = 1/2$. The V_{600} closure programme has established, in eight findings [1], a substrate-side framework in which: the critical line embeds into V_{600} via the Hilbert modular L-function $L_{\text{sub}} = \zeta_K \cdot \zeta_K(s-1) \cdot C_2$; the closure operator C_φ has a Galois decomposition $94 + 13 + 13$; and the witness invariant $W = V_{\text{min}}$ is the universal attractor of closure-flow [4].

This paper states a *conditional localisation* of the Hilbert–Pólya problem onto that framework.

Proposition (one direction, classical). *If a self-adjoint operator T_W exists on an infinite-dimensional lift $\mathcal{H}_+$ of the τ -fixed block of V_{600} , satisfying the substrate constraints of Definition 2, whose positive eigenvalues are in multiplicity-respecting bijection with the non-trivial zeros of ζ in the upper half plane (via $\rho = \frac{1}{2} + it_\rho$; the conjugate lower half follows by symmetry), then RH holds.*

This direction is the classical Hilbert–Pólya implication (self-adjointness forces a real spectrum) and carries no novelty; what is ours is the *constraint list* — ground state at V_{min} , commutation with closure-flow and with the Hecke action of the Hilbert modular lift — which localises *where* such an operator would have to live. Three honesty notes, up front. First, no lift $\mathcal{H}_+$ is constructed here: the τ -fixed block is 94-dimensional, a finite space cannot carry the infinite spectrum, and producing a lift with the stated properties is the open problem this paper names. Second, the converse — that RH guarantees a generator satisfying the substrate constraints — is *not* established here, and without the constraints the converse is trivial (a diagonal operator on ℓ^2 does it), so the entire content of the localisation lies in the constraints. Third, this is a specification, not a theorem about ζ : it sharpens the question, it does not move the answer.

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1 Why RH is the final reframe

The V_{600} closure programme has produced a structural picture in which the Riemann critical line has a precise substrate interpretation. The Witness Invariant Theorem [4] establishes:

- the substrate is the 600-cell vertex set V_{600} ;
- the closure equation $W = \Phi \times G \times O$ subject to $C \rightarrow 0$ has the unique solution $W = V_{\min}$;
- every closure-flow trajectory attracts to $V_{\min}$;
- the critical line embeds into the τ -fixed 94 block via L_{sub} .

What remains open is the analytical step: are the ordinates γ_n along the critical line $s = 1/2 + it$ *forced* to lie at the eigenvalues of some specific operator on the substrate’s τ -fixed block? This is the substrate-side version of the Hilbert–Pólya conjecture.

1.1 The role of RH in the puzzle

The Witness Invariant Theorem identifies *where* the witness lives ($V_{\min}$). It does not specify *which excited eigenvalues* the substrate’s spectral generator has. RH is the statement that constrains these eigenvalues to be exactly $\{\gamma_n\}$.

In the diagram’s vocabulary ([4], Fig. 1):

- “balance 1/2 holds” — this is the witness axis $\text{Re}(s) = 1/2$, fixed by the functional equation;
- “ t unfolds” — the spectral parameter t along the critical line is the parameter that gives the spectral excitation;
- the ordinates γ_n are the t -values where ζ vanishes on the line (under RH, all of them).

RH says these t -values are real and lie at specific positions. The substrate framework identifies $V_{\min}$ as “ $t = 0$, balance” and asks whether the excited eigenstates have t -values matching the γ_n .

1.2 What this paper does

We articulate the precise sense in which a substrate-localised Hilbert–Pólya operator would witness RH. The main statement (§3) is a *conditional localisation*: the existence of such an operator — on a lift satisfying the substrate constraints, its positive eigenvalues in multiplicity-respecting bijection with the upper-half-plane non-trivial zeros — implies RH, by the classical argument. The converse is stated only as the target it is: an open construction problem. We do not prove RH, and we do not claim an equivalence.

2 The substrate framework, briefly

2.1 Setup

- V_{600} is the 600-cell vertex set, $|V_{600}| = 120$, identified with the binary icosahedral group $2I$.
- A_1 is the edge adjacency operator on V_{600} .
- $C_\varphi = (12 + \varphi^{-2})I - A_1$ is the closure operator.
- C_φ has nine eigenvalues in $\mathbb{Z}[\varphi]$ with multiplicities summing to 120.
- The Galois action $\sigma : \varphi \mapsto 1 - \varphi$ partitions the spectrum into 94 rational (the τ -fixed block V_+) and 26 irrational (the τ -paired block V_- , decomposing further as $13 + 13$).
- $V_{\min} = \ker(C_\varphi - \varphi^{-2}I)$ is 1-dimensional and sits inside V_+ .

2.2 The substrate L-function

The icosian theta L-function [2]:

$$L_{\text{sub}}(s) = \zeta_K(s) \cdot \zeta_K(s-1) \cdot C_2(s), \quad K = \mathbb{Q}(\sqrt{5}),$$

with $\zeta_K(s) = \zeta(s) \cdot L(s, \chi_5)$ and $C_2(s) = 1 - 2 \cdot 2^{-s} + 2^{2-2s}$.

2.3 The closure equation

The triadic closure equation [4]:

$$W = \Phi \times G \times O \quad \text{subject to} \quad C \rightarrow 0$$

has the unique solution $W = V_{\min}$, where:

- Φ (Phase) = closure-flow dynamics on $\mathbb{C}^{120}$,
- G (Geometry) = V_{600} with A_1 ,
- O (Observer) = observer-instance trajectories,
- $C = \|P_- v\|^2 \rightarrow 0$.

3 The localisation target

3.1 The spectral generator and the witness axis

Definition 1 (Substrate lift). *A substrate lift of the τ -fixed block is a separable infinite-dimensional Hilbert space $\mathcal{H}_+$ together with an isometric embedding $V_+ \hookrightarrow \mathcal{H}_+$ and extensions of the closure-flow and of the Hecke action of the Hilbert modular lift of V_+ to $\mathcal{H}_+$. No such lift is constructed in this paper; exhibiting one with the properties below is the open problem this paper specifies. (The finite block V_+ itself is 94-dimensional and cannot carry the infinite spectrum required below — the lift is not optional.)*

Definition 2 (Substrate-localised generator). *A substrate-localised generator for the witness invariant $W = V_{\min}$ is a self-adjoint operator T_W on a substrate lift $\mathcal{H}_+$ (Definition 1) such that:*

- (i) $V_{\min}$ is the ground state of T_W (smallest eigenvalue, eigenvalue normalised to 0 for definiteness);
- (ii) T_W commutes with the closure-flow on V_+ ;
- (iii) T_W commutes with all Hecke operators acting on the Hilbert modular form lift of V_+ ;
- (iv) the positive spectrum of T_W is discrete (eigenvalues counted with multiplicity; no simplicity is assumed — simplicity of the ζ -zeros is itself open).

Definition 3 (Positive spectrum). *For a substrate-localised generator T_W on $\mathcal{H}_+$, write*

$$\mathcal{R}(T_W) = \{t \in \mathbb{R}_+ : \exists v \in \mathcal{H}_+, v \perp V_{\min}, T_W v = t \cdot v\}$$

for its set of strictly positive eigenvalues; whenever this set is compared with a zero multiset, the comparison is by multiplicity-respecting bijection (eigenvalue multiplicities against zero multiplicities), not bare set equality. (In v1 of this paper this set carried a decorative name; it is simply the positive spectrum, and we call it that.)

3.2 The conditional statement

Proposition 4 (Conditional localisation: the classical direction). *Let T_W be a substrate-localised generator (Definition 2) on some substrate lift $\mathcal{H}_+$, and suppose there is a bijection, respecting multiplicities, between the non-trivial zeros ρ of ζ with $\text{Im}(\rho) > 0$ and the positive eigenvalues t_ρ of T_W , such that $\rho = \frac{1}{2} + it_\rho$ for every such zero. Then RH holds. (The zeros with $\text{Im}(\rho) < 0$ are determined by the conjugation symmetry $\rho \mapsto \bar{\rho}$, so the upper-half-plane set suffices.)*

Proof. Self-adjointness of T_W forces every eigenvalue t_ρ to be real, so every upper-half-plane zero satisfies $\rho = \frac{1}{2} + it_\rho$ with $t_\rho \in \mathbb{R}$, i.e. $\text{Re}(\rho) = \frac{1}{2}$; the conjugation symmetry extends this to all non-trivial zeros. This is the classical Hilbert–Pólya implication; nothing about the substrate is used in the deduction, and we claim no novelty for it. $\square$

Target 5 (The substrate localisation target). *Construct a substrate lift $\mathcal{H}_+$ and a substrate-localised generator T_W on it with $\mathcal{R}(T_W) = \{\gamma_n\}$ as in Proposition 4. The content of the target is entirely in the substrate constraints (ground state $V_{\min}$, closure-flow and Hecke commutation): without them the statement “RH implies such an operator exists” is trivially true (under RH, take the diagonal operator on ℓ^2 whose eigenvalues are the upper-half-plane ordinates, repeated with multiplicity), and with them it is open. We do not assert that RH guarantees a generator satisfying the constraints; that assertion was the central defect of v1 of this paper and is withdrawn.*

Remark 6 (On the withdrawn equivalence). *v1 stated an equivalence between RH and “substrate spectral completeness”. One direction is Proposition 4; the other was supported only by the assumption that the construction can be carried out under RH, which assumes what it must prove. A third clause (“the closure equation is spectrally complete”) was definitional — equivalent to the first by unwinding the definitions — and is retained only as terminology, not as a result. An analogous conditional statement can be formulated for $L(s, \chi_5)$ on the τ -paired block V_- ; it has the same one-direction status and the same open converse, and we state it below only as a conjectural target.*

3.3 What the localisation says precisely

The localisation, stated precisely

The statement of this paper, in one breath:

If the witness invariant $W = V_{\min}$ admits a substrate-localised generator on some lift of the τ -fixed block whose positive eigenvalues are in multiplicity-respecting bijection with ζ ’s upper-half-plane non-trivial zeros (via $\rho = \frac{1}{2} + it_\rho$; the conjugate lower half follows by symmetry), then RH holds. Whether RH conversely guarantees a generator satisfying the substrate constraints is open, and is the localisation target.

This is not a proof of RH, and not an equivalence. Closing the target requires either:

- **Bottom-up:** explicitly constructing T_W from the substrate’s intrinsic data (closure-flow + cuspidal Hilbert modular forms over $\mathbb{Q}(\sqrt{5}) + 2I$ -equivariance). This is the Hilbert–Pólya programme applied to V_{600} .
- **Top-down:** proving that any Hilbert–Pólya operator on the relevant Hilbert modular form space must localise onto $V_{\min}$ ’s span (the τ -fixed block); then T_W exists by analytic construction, and RH follows.

The first resolution would prove RH (Proposition 4); the second would close the localisation target but still leaves the construction to do. The substrate framework sharpens where the operator must live; it does not move the answer.

3.4 Extension: $L(s, \chi_5)$ on the τ -paired block

Target 7 (τ -paired analogue, conjectural). *The same one-direction statement can be formulated for $L(s, \chi_5)$ on a lift of the τ -paired block V_- : a substrate-localised generator there whose positive eigenvalues are in multiplicity-respecting bijection with the non-trivial zeros of $L(s, \chi_5)$ in the upper half-plane (via $\rho = \frac{1}{2} + it_\rho$; the conjugate-symmetric lower half follows) would imply GRH_{χ_5} by the identical self-adjointness argument. The Dedekind factorisation $\zeta_K = \zeta \cdot L(s, \chi_5)$ motivates the block assignment. No converse is asserted, and no construction is given; this is a target, stated for symmetry.*

3.5 Corollaries

Remark 8 (Combined conditional statement). *If substrate-localised generators exist on lifts of both blocks with the stated spectra, then RH and GRH_{χ_5} both hold, by applying Proposition 4 and Target 7’s implication to each factor of $\zeta_K = \zeta \cdot L(s, \chi_5)$. (The finite direct sum $V_+ \oplus V_- = \mathbb{C}^{120}$ of $v1$ is replaced by the direct sum of the lifts; the finite space cannot carry either spectrum.)*

4 What this reframe accomplishes and what remains open

4.1 What the reframe accomplishes

1. **A constraint list for Hilbert–Pólya, anchored in the substrate.** Any operator meeting Definition 2 — ground state at $V_{\min}$, commutation with closure-flow and with the Hecke action of the Hilbert modular lift — would witness RH by Proposition 4. The constraints are checkable; the construction is not supplied.
2. **An honest size statement.** The τ -fixed block itself is 94-dimensional, so the search is *not* finite-dimensional ($v1$ claimed it was; that claim is withdrawn): any candidate lives on an infinite-dimensional lift anchored to the block.
3. **A precise interface to the closure equation.** The closure equation $W = \Phi \times G \times O$ subject to $C \rightarrow 0$ acquires a definite operator-side reading — as a *definition* of what its spectral completeness would mean, not as a validated theorem.

4.2 What remains open

1. **Constructing T_W from substrate-intrinsic data.** The translation engine has shown [3] that finite-group operators on V_{600} alone do not produce $\{\gamma_n\}$; the missing ingredient is cuspidal Hilbert modular form data over $\mathbb{Q}(\sqrt{5})$. Once such data is available (via LMFDB, Magma, or future SAGE), the construction can be attempted.
2. **The classical Hilbert–Pólya conjecture.** Independent of the substrate’s localisation, the existence of a self-adjoint Hilbert–Pólya operator for ζ is open. The substrate framework restricts the search domain but does not produce the operator.
3. **Substrate-level positivity arguments.** A direct substrate-side proof would need to show that any substrate-localised generator on a lift of V_+ *must* have its spectrum on $\mathbb{R}_+$. The current substrate framework permits this as a possibility but does not prove it.

Scope

What is established.

- One direction (Proposition 4): a substrate-localised generator on a lift, its positive eigenvalues in multiplicity-respecting bijection with the upper-half-plane non-trivial zeros, would imply RH — by the classical self-adjointness argument, with no novelty claimed for the deduction.
- A precise constraint list (Definition 2) localising where such an operator must live.

What is not established.

- The Riemann Hypothesis, GRH_{χ_5} , or any analytic statement about ζ or $L(s, \chi_5)$.
- The converse: that RH guarantees a generator satisfying the substrate constraints (v1’s claimed equivalence, withdrawn).
- The existence of a substrate lift, or of T_W on any lift.
- Any new computation: this bundle contains no simulations; all structural counts (120, 94+13+13, $\dim V_{\min} = 1$) are imported from the cited sibling artifacts (`critical-line-pullback`, `the-24-600-spectral-bridge`).

5 Conclusion

The Riemann Hypothesis is localised, not resolved, and not reformulated as an equivalence. The substrate framework specifies *where* a Hilbert–Pólya operator for ζ would have to live ($V_{\min}$ -anchored, on an infinite-dimensional lift of the τ -fixed block) and *what* structural properties it would have to satisfy (Definition 2); its existence with the stated spectrum would imply RH (Proposition 4), by the classical argument. Whether RH conversely guarantees an operator meeting the substrate constraints is open — that is the localisation target, and the honest name for what this paper contributes is a *specification*. The analytical step that would close it is the same one that has been open since Riemann.

Acknowledgements. This paper synthesises the programme’s preceding work; no new techniques are introduced. The structural diagram licensed by the closure equation $W = \Phi \times G \times O$ subject to $C \rightarrow 0$ is due to the *VFD Witness Structure* framework [4].

References

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